SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – SHORT ANSWER TEST

Course Code:	CSA0309	Course Title:	Data Structures
CO Assessed:	CO5 – Manipulation (BL3, BL4)	Assessment:	Assessment Tool 3 – Short Answer Test
Weightage:	30% of CIA	Total Marks:	100
CO2 Statement:	*Construct MST using shortest path algorithms and traverse the graph to model and analyse real-world problem.*		
Student Name:	Bhuvaneshwaran.J	Reg. No.:	192511402
Section / Batch:	Slot B	Date:	02/09/2026

Code of Conduct

I Bhuvaneshwaran.J (192511402) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that ~~any~~ violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: ________

Instructions

- This test contains 10 short answer test questions. Total: 100 Marks.
- When a student answer using an Analytical problem-solving, in evaluation the answer should be with following five requirements: Understanding of problem, select appropriate data structure, Design the solution, analyze, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.
- No DTP printouts or electronic devices permitted. They should provide written materials.

Section / Topic	Focus	Q Nos	Marks
Minimum Spanning Tree	Kruskal's Algorithm	1	20
Graph Sorting	Topological Sort	2	20
Shortest Path Algorithm	MST & Dijkstra's Algorithm	3	20
Shortest Path Algorithm	Dijkstra's Algorithm	4	20
Minimum Spanning Tree	Prim's or Kruskal's Algorithm	5	20
	GRAND TOTAL:	100 Marks	

Annexure A - Short Answer Test

1. Perform the following operations in the Singly Linked List for the following input.
   Input: 22-> 77-> 55-> 88->
   i. Insert a new-node 110 at the end of the list.
   ii. Insert a new-node 101 at the position 4 of the list.
   iii. Delete a node at the beginning of the single linked list
   iv. Delete a node at the position 3 of the single linked list.
2. Rotations in each step of insertion. 1,2,3,4,5,6,7,8,9,10 (), 4) Push (90), 5) Push (36), 6) Push (11), 7) Push (88), 8) Pop (), 9) Pop (). Draw the diagram of the stack and illustrate the above operations and identify where the top is?
3. Convert the following infix into a postfix expression. → B - (C/D + (E % F* G) / H) * J
4. Evaluate the given postfix expression. 53+62/*35*+
5. Construct a B-Tree of Order 3 by inserting numbers 34,26,45,12,87,56,78,22,23,24,65,85,95.
6. Construct a 2-3-4 Tree by inserting the following sequence of numbers 45, 36, 76, 23, 89, 115, 98, 39, 41, 56, 69, 48.
7. Build a max heap for the following numbers - 20, 35, 23, 22, 4, 45, 21, 5, 42 and 19. At each stage of heap tree construction, ensure heap properties are satisfied.
8. Perform the operations onto splay tree (draw after each operation): insert (3), insert (9), insert (12), insert (55), insert (1), insert (3), delete (3), delete (9), delete (60), insert (3), search (12).
9. Describe the algorithm to sort the elements 170, 45, 75, 90, 802, 24, 2, 66 using Radix sort. Explain with each phase?
10. Consider the following graph & construct BFS.

Graph with six nodes

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Conceptual Understanding	30	Demonstrates complete and accurate understanding of concepts	Good understanding with minor gaps	Basic understanding	Limited understanding
Accuracy of Answer	25	Answers are completely correct and relevant	Mostly correct with minor errors	Partially correct	Mostly incorrect
Problem-Solving & Reasoning	20	Provides logical reasoning and appropriate steps	Good reasoning with minor gaps	Basic reasoning	Lacks logical reasoning
Clarity & Relevance	15	Answers are precise, clear, concise, and directly address the question	Mostly clear and relevant	Somewhat unclear or incomplete	Irrelevant or unclear answers
Time Management & CompletionS	10	Completes all questions accurately within the time	Completes most questions on time	Requires additional time	Unable to complete the test

SFaculty In-charge

Signature: ____________

Date: ____________

Course Coordinator

Signature: ____________

Date: ____________

Program Director

Signature: ____________

Date: ____________

Perform the operations in the singly linked list

Initial list:

22 → 77 → 55 → 88

(i) insert 110 at the end:

22 → 77 → 55 → 88 → 110

(ii) Insert 101 at position 4

22 → 77 → 55 → 101 → 88 → 110

(iii) Delete node at the beginning

Delete 22

77 → 55 → 101 → 88 → 110

(iv) Delete node at position 3

Final list:

77 → 55 → 88 → 110

2 Rotation in each step of insertion (1, 2, 3, 4, 5, 6, 7, 8, 9, 10)

operations	Stack	Top
Push (90)	90	90
Push (36)	90, 36	36
Push (11)	90, 36, 11	11
Push (88)	90, 36, 11, 88	88
Pop ()	90, 36, 11	11
Pop	90, 36	36

(3) Convert the infix into postfix

B - (C / D + E % F * G) / H)

Postfix expression

BCD / EF % G * H / + J * -

without spaces

BCD/EF%G*H/+J*-

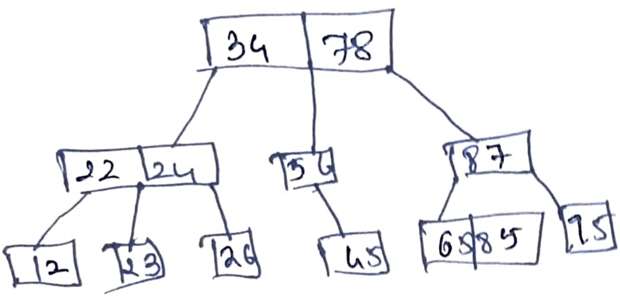
(4) Evaluate the given postfix expression 53+62/*35*+

Token	Operations	Stack
5	Push	5
3	Push	5,3
+	5+3	8
6	Push	8,6
2	Push	8,6,2
/	6/2	8,3
*	8×3	24
3	Push	24,3
5	Push	24,3,5
*	3×5	24,15
+	24+15	39

The answer is 39

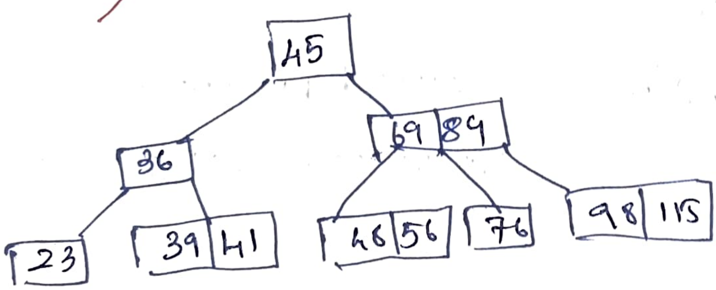
Construct a B tree of order 3 34, 26, 45, 12, 87, 56, 78, 22,
24, 65, 85, 95

For an order of 3 Btree, a node can contain at most 2 keys



⑥ Construct a 2-3-4 Tree 45, 36, 76, 23, 89, 115, 98, 39, 41, 56
69, 48

2-3-4 Tree



⑦ Build a max heap

Numbers: 20, 35, 23, 22, 4, 45, 21, 5, 42, 19

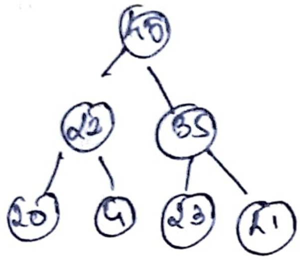
(1) 20

(2)

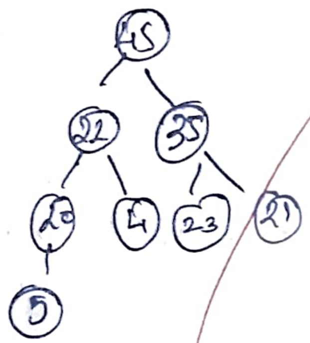
(3)

(4)

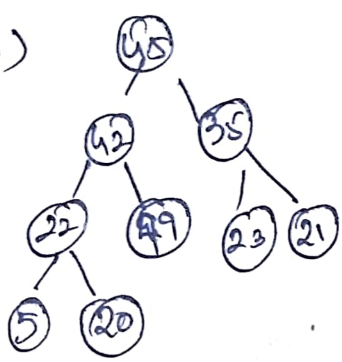
(5)



(6)



(7)



Array Representation:

[45;42;35,22,19,21,5,20,4]

(8) Perform the operations onto splay tree

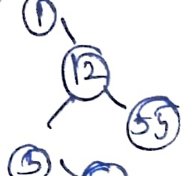
(1) 3

(2)

(3)

(4)

(5)

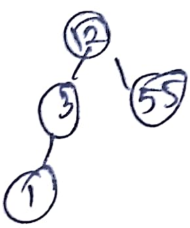


6)

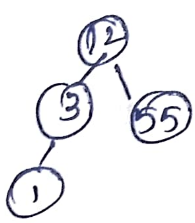
(7)

(8)

(9)



Final splay tree:



Describe the algorithm to sort elements
170, 45, 75, 90, 802, 24, 2, 66

Phase 1 unit digit

170, 907, 802, 2, 24, 45, 75, 66

Phase 2 Tens Digit

802, 2, 24, 45, 60, 170, 75, 90

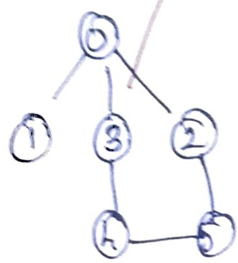
Phase 3- Hundred digit

2, 24, 45, 66, 75, 90, 170, 802

Final Sorted:

2, 24, 45, 66, 75, 90, 179, 802.

(Q) Consider the following graph & construct BFS



Step 1:

Start at 0

Queue: 0 , visited: 0

Step 2:

visit neighbours:

1, 2, 3

Queue: 1, 3, 2

Step 3: Process 1 ; No new vertex

Step 4: Process 2 ; No new vertex

Step 5: Process 4 ; visit 5

BFS Tree